HARD WATER 18

When soap solution is added to soft water and shaken, a lather (foam) readily forms. If soap solution is added to hard water, lather does not form immediately. More soap has to be added until it removes all of the hardness. Hardness in water wastes soap. Other indications of hard water include hard deposits in hot water systems and kettles. Synthetic ('soapless') detergents are not affected by hard water.

There are two sorts of hardness: temporary and permanent. Temporary hardness is caused by carbonates and can be removed by boiling. Permanent hardness is not removed by boiling. Chemicals called *water softeners* can be used to remove hardness so that less soap is needed. Dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions are a major source of hardness. Water can also be 'softened' by passing it through an ion exchange medium, where calcium and magnesium ions are replaced by sodium ions. The medium is usually made up of organic resins and can be regenerated by passing brine (salt solution) through it. Chemical water softeners work in a different way; they react with these ions and remove them from solution by precipitating them out.

A student carried out a series of tests to compare two water softeners: washing soda (sodium carbonate decahydrate, $Na_2CO_3.10H_2O$) and a commercial brand, *Softa*. Below is an extract from her practical report.

Method 50 mL of hard water was placed into a conical flask. 1.0 mL of soap solution was added to the flask, which was then stoppered and shaken. More soap solution was added, 1.0 mL at a time and shaking repeated until a permanent lather formed. The total volume of soap solution used was recorded.

This procedure to obtain a lather was repeated, but 0.1 g of sodium carbonate was added first. This was done with other quantities of sodium carbonate and also with equivalent amounts of *Softa*.

Results

Mass of softener (g)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7
Volume of soap using sodium carbonate (mL)	24.0	16.0	8.0	3.0	3.0	3.0	3.0	3.0
Volume of soap using *Softa* (mL)	24.0	19.0	14.0	9.0	5.0	1.0	1.0	1.0

1 Name the piece of apparatus that should be used to measure out each of the following.

- **a** the hard water sample
- **b** the soap solution

2 Outline how the student ensured that this procedure was a 'fair test'.

3 On the same set of axes, draw a graph of volume of soap solution (y-axis) against mass of softener (x-axis) for each softener.

4 Referring to your graph from Q3, discuss the relative effectiveness of the two chemicals as water softeners.

5 Name one way of avoiding the problems of using soap rather than softening the water.

6 Describe some indications that a domestic water system has hard water.

7 Outline the difference between temporary and permanent hardness.

8 *When ions from two soluble compounds react to form an insoluble substance.* Which chemical phenomenon does this describe?

9 Distillation can be used to remove hardness from water but is not used on a large scale. Give a reason why distillation isn't used to purify water in large quantities.

10 What is *brine*?

Research **11** Find out the actual quantities of dissolved solids required for water to be classified as *hard*.

12 Hard water does have some advantages. Find out two of these advantages.

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